Anfrage Kabelablängen
Fa. Metzner

Modulare Automation

Gluth SYSTEM TECHNIK

Aufgabenstellung

Herstellen einer Anlage zur automatischen Konfektionierung und Prüfung von Kabeln.

Auszüge aus dem Lastenheft:

Designation	Amount	Material number
Cable (Low Voltage singlecore cable 16mm²/25mm² /35mm² FLRY-B;FLR2G;) 	1 pc	L0263CXX (16mm²) L3363066 (16mm²) L3366063 (25mm²) L3369048 (35mm²)

Dimensions [mm] (Cable Length)	300 - 1500

15.4 Cutting to length

1. Cable entry
   - Reliably inputting of the cable, cable transport mustn´t be hindered by machine parts
   - Conical straightening device, declining with at least 4 + 5 rolls, scale and quick-fastener
   - Cable feeding: min 2,5m / sec, freely programmable
   - Acceleration freely programmable

   - Length repeat accuracy:
     - Cable length smaller 500mm < +/-1mm
     - Cable length >=500mm < 0,2%
   - Observing the + tolerance area has to be adjustable by correction factor
   - Mechanical knot control (diameter specific and marked adjusting disc)
   - Minimum cable waste (1. Cable, zero cut)

2. Cut-in units
   - Pre-switching of a cable roller with cable store, types or power specified by the consultant
   - Interface description cut-in units
   - Automatic identification if the actual stripping blade is the correct one for the loaded order.

3. Length cutting
   - Length area: 500 - 2000mm
   - Cutting blades
     - Blade quality: standard materials and options, tolerances
     - Lifetime for each type

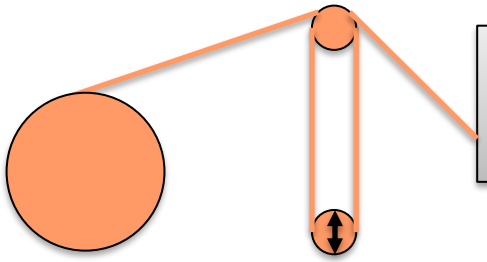
15.5 Stripping

1. Stripping length of each individual line 2 - 20mm, tolerance +/- 0.2mm;
   - Freely adjustable
   - Adjustable in 0,1mm steps
2. The stripping length is transferred via MES including tolerance.
3. Knifes (specified from the supplier)
   - Stripping knifes
   - Gradation based on the wire dimensions
   - Types: V-blades, radius-V-blades, radius-blades
4. Blade quality: standard material and options
5. Lifetime per model
6. Integrated vacuum cleaning
   - No clogging of the suction hoses
   - Collection container must be designed in such a way that the container only has to be emptied once per shift (8 hours)
   - Collection container with quick change system
   - 2 collection containers should be offered

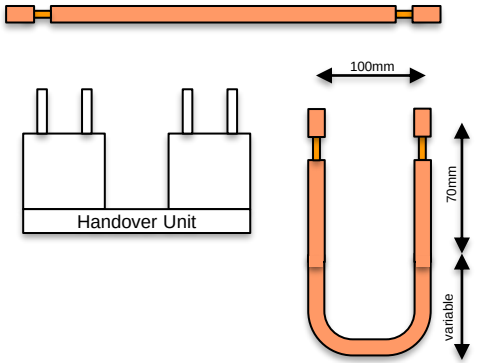
Hinweis. Verschiedene Angaben im Lastenheft

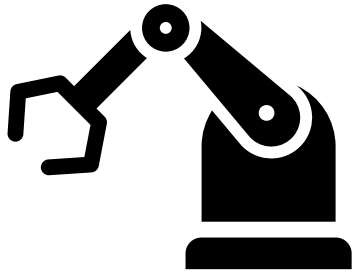
Gluth SYSTEM TECHNIK

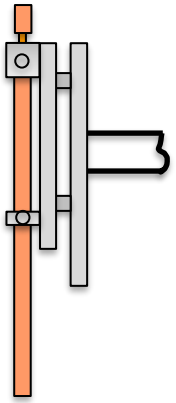
Konzept Kabelzuführung



Cut & Strip







Feeding Unit	Cut & Strip Unit	Handover Unit	Robot Handling	Round Table

Lieferant ←→ ???? ←→ Gluth

Rahmendaten

Taktzeit: 6 Sekunden
Übergeordnete Steuerung: Siemens
Vorgabe der Kabelparameter durch die übergeordnete Steuerung

Teilabzug beim Abisolieren geplant
„Handover Unit“ soll das Kabel übernehmen und in U-Form bringen (Klärung ob Lieferant oder Gluth)
Roboter übernimmt das Kabel und übergibt es an den Rundtisch (Gluth)

Offene Punkte:

- Anlieferung der Kabel (Größe der Spule, Innendurchmesser, Gewicht)